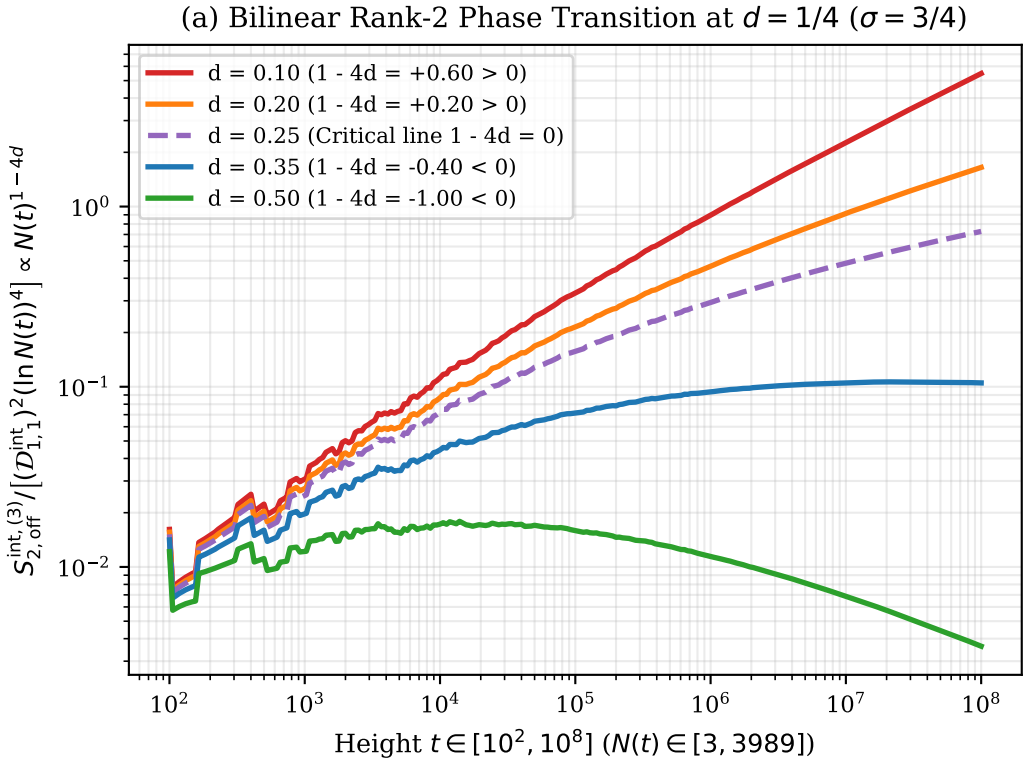
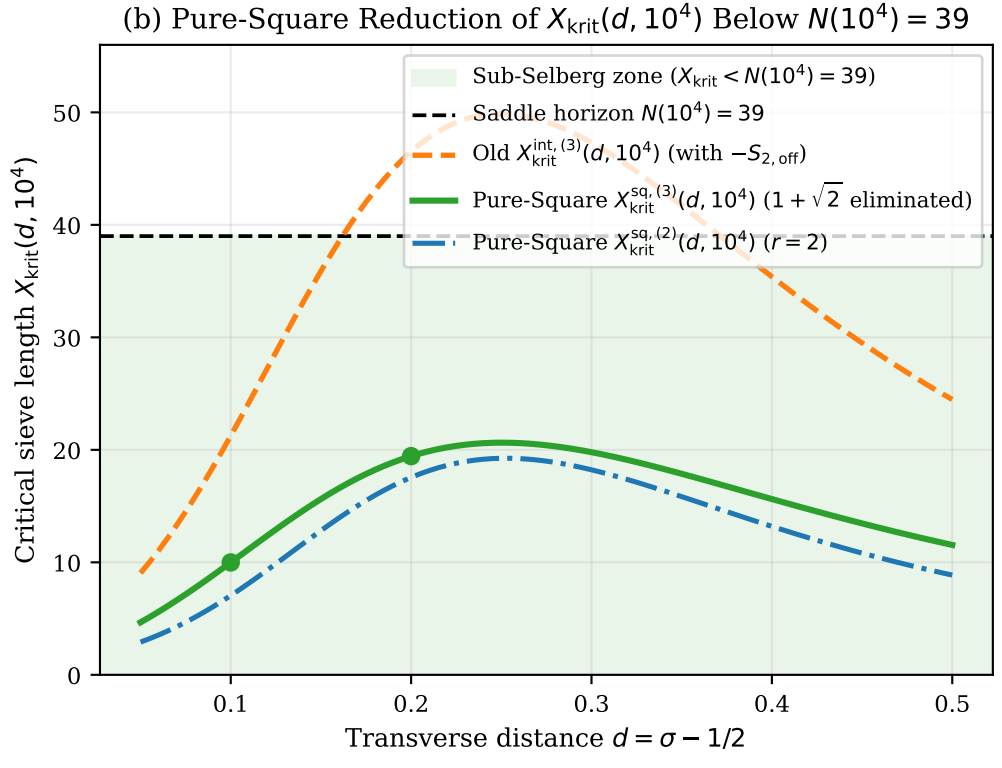
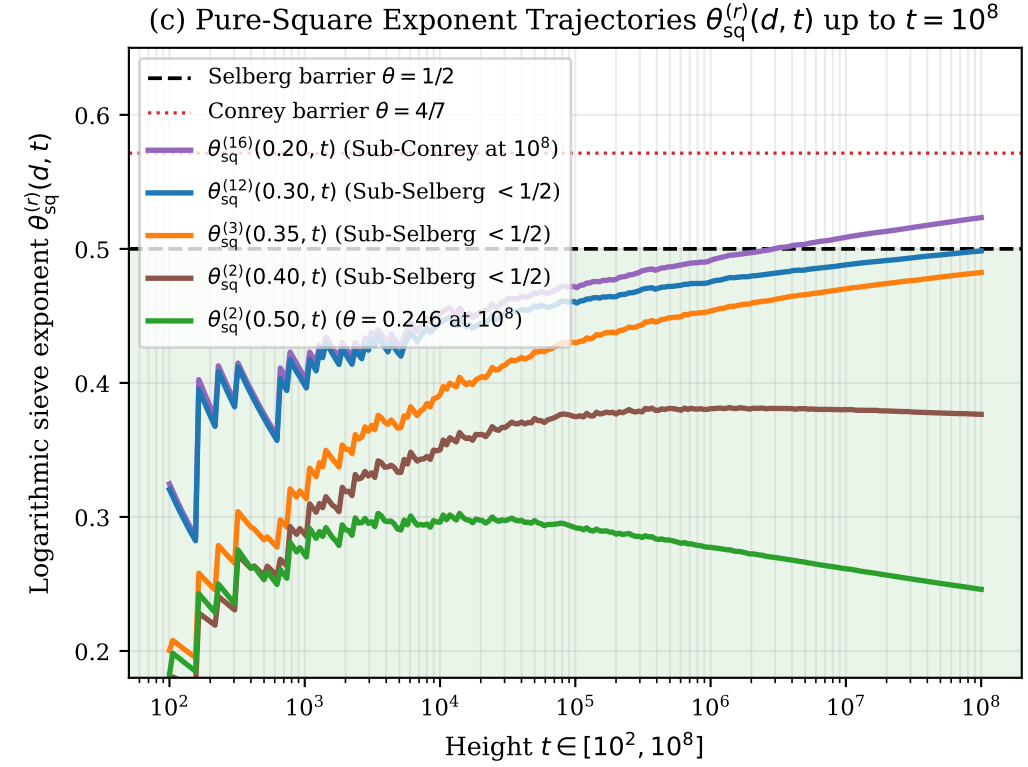
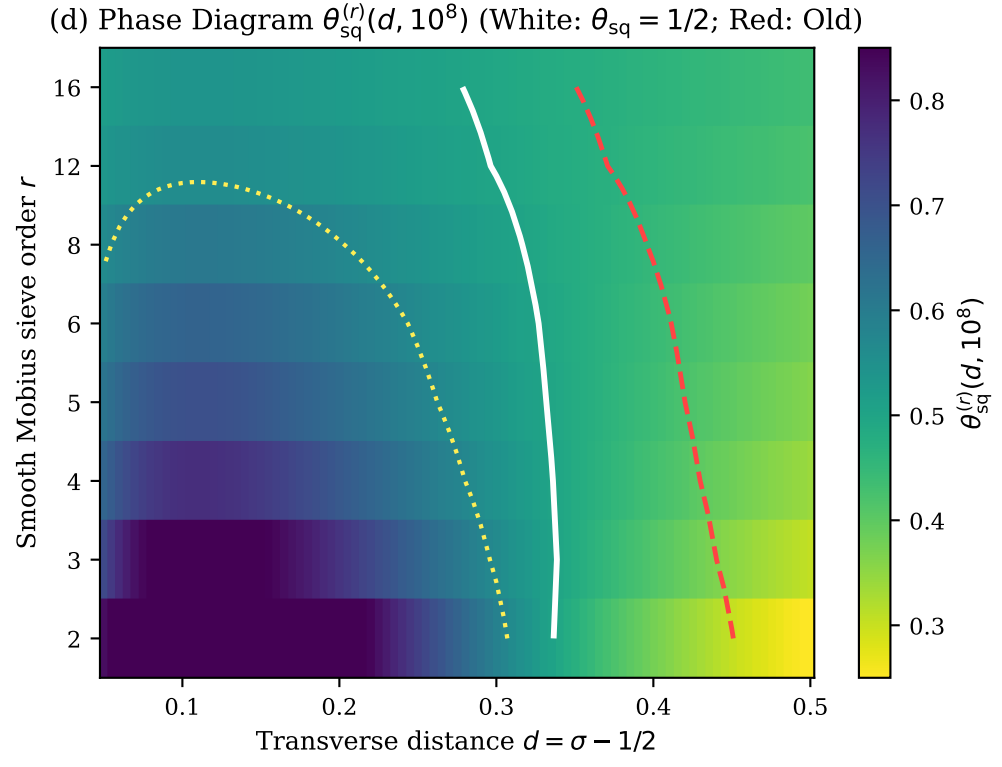

(a) Bilinear Rank-2 Phase Transition at $d = 1/4$ ($\sigma = 3/4$)
d = 0.10 (1 - 4d = +0.60 > 0)
d = 0.20 (1 - 4d = +0.20 > 0)
d = 0.25 (Critical line 1 - 4d = 0)
d = 0.35 (1 - 4d = -0.40 < 0)
d = 0.50 (1 - 4d = -1.00 < 0)
$S^{\mathrm{int},(3)}_{2,\mathrm{off}} / [(D^{\mathrm{int}}_{1,1})^2 (\ln N(t))^4] \propto N(t)^{1-4d}$
Height $t \in [10^2, 10^8]$ ($N(t) \in [3, 3989]$)


(b) Pure-Square Reduction of $X_{\mathrm{krit}}(d, 10^4)$ Below $N(10^4) = 39$
Sub-Selberg zone ($X_{\mathrm{krit}} < N(10^4) = 39$)
Saddle horizon $N(10^4) = 39$
Old $X^{\mathrm{int},(3)}_{\mathrm{krit}}(d, 10^4)$ (with $-S_{2,\mathrm{off}}$)
Pure-Square $X^{\mathrm{sq},(3)}_{\mathrm{krit}}(d, 10^4)$ ($1 + \sqrt{2}$ eliminated)
Pure-Square $X^{\mathrm{sq},(2)}_{\mathrm{krit}}(d, 10^4)$ ($r = 2$)
Critical sieve length $X_{\mathrm{krit}}(d, 10^4)$
Transverse distance $d = \sigma - 1/2$


(c) Pure-Square Exponent Trajectories $\theta^{(r)}_{\mathrm{sq}}(d, t)$ up to $t = 10^8$
Selberg barrier $\theta = 1/2$
Conrey barrier $\theta = 4/7$
$\theta^{(16)}_{\mathrm{sq}}(0.20, t)$ (Sub-Conrey at 10^8)
$\theta^{(12)}_{\mathrm{sq}}(0.30, t)$ (Sub-Selberg $< 1/2$)
$\theta^{(3)}_{\mathrm{sq}}(0.35, t)$ (Sub-Selberg $< 1/2$)
$\theta^{(2)}_{\mathrm{sq}}(0.40, t)$ (Sub-Selberg $< 1/2$)
$\theta^{(2)}_{\mathrm{sq}}(0.50, t)$ ($\theta = 0.246$ at 10^8)
Logarithmic sieve exponent $\theta^{(r)}_{\mathrm{sq}}(d, t)$
Height $t \in [10^2, 10^8]$


(d) Phase Diagram $\theta^{(r)}_{\mathrm{sq}}(d, 10^8)$ (White: $\theta_{\mathrm{sq}} = 1/2$; Red: Old)
Smooth Mobius sieve order r
Transverse distance $d = \sigma - 1/2$
$\theta^{(r)}_{\mathrm{sq}}(d, 10^8)$